

Bubble ROI Visualization Bug Fix Report

2D-RAFT-DIC-GUI | Per-Frame ROI System

Root Cause Analysis

Bug 1 – Reference Mode: ROI shape changes per frame

- `_apply_current_frame_mask()` uses `np.round(x + U)` to check deformed positions
- Near the bubble boundary, rounding pushes coordinates inside the bubble
- These pixels are set to NaN and baked into `session.displacement_results`
- Reference-mode rendering clips to frame 0's mask, but cannot recover the NaN
- Result: visible ROI shrinks as bubble grows (more edge pixels get NaN)

Bug 2 – Deformed Mode: ROI stops following mask during bubble shrink

- `compute_inverse_map()` used forward-mapped contour for output bounding box
- When bubble shrinks, new material appears beyond the contour extent
- Bounding box was too small → pixels outside it were clipped

Fix Applied

Bug 1: `fill_mask_nan_gaps()` – nearest-neighbor interpolation

- After clipping to frame 0's mask, identify "gap" pixels:
`mask[pixel] == True BUT data[pixel] == NaN`
- Fill each gap pixel with the nearest valid pixel's value
- Uses `scipy.ndimage.distance_transform_edt` (O(n), very fast)
- Result: visible ROI shape exactly matches frame 0's mask for ALL frames

Bug 2: Bounding box expansion in `compute_inverse_map()`

- When a user mask is provided, expand output bounding box to include the full extent of the user mask (not just the forward-mapped contour)
- User mask replaces contour as the authoritative validity boundary
- Result: deformed-mode ROI follows user mask during both growth and shrink

Test Scenario

- 200×200 px image, bubble center at (100, 100)
- Bubble radii: [12, 16, 20, 25, 30, 26, 22, 18, 14] pixels
- Incompressible 2D plane-strain: $r' = \sqrt{r^2 + R^2 - R_0^2}$
- Two variants: clean (no noise) and noisy ($\sigma = 2.0$ px near edge)
- `_apply_current_frame_mask` simulated with same `np.round()` logic

Quantitative Results — Reference Mode Pixel Counts

Clean Displacement (no noise)

Frame	0	1	2	3	4	5	6	7
Bubble R	16	20	25	30	26	22	18	14
Before fix	39271	38727	38131	37131	37811	38451	38827	39551
After fix	39559	39559	39559	39559	39559	39559	39559	39559
Expected	39559	39559	39559	39559	39559	39559	39559	39559
Deficit	288	832	1428	2428	1748	1108	732	8

Noisy Displacement (noise $\sigma = 2.0$ px)

Frame	0	1	2	3	4	5	6	7
Bubble R	16	20	25	30	26	22	18	14
Before fix	39197	38661	38004	37003	37694	38359	38765	39473
After fix	39559	39559	39559	39559	39559	39559	39559	39559
Expected	39559	39559	39559	39559	39559	39559	39559	39559
Deficit (before)	362	898	1555	2556	1865	1200	794	86

Reference Mode — Before vs After Fix (Clean Data)

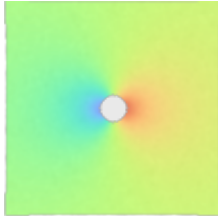
R=16
39271px



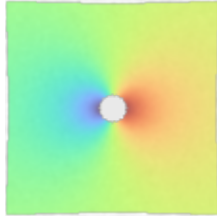
R=20
38727px



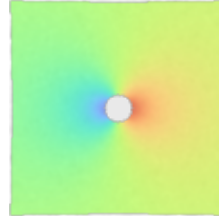
R=25
38131px



R=30
37131px



R=26
37811px



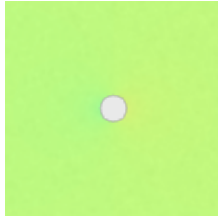
R=22
38451px



R=18
38827px



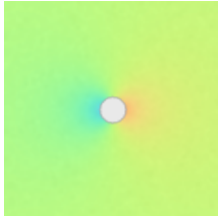
R=14
39551px



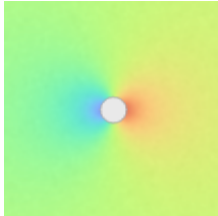
39559px



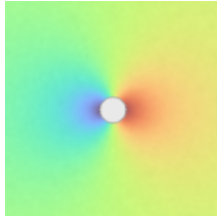
39559px



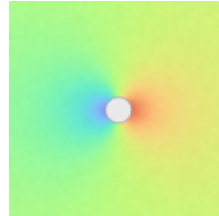
39559px



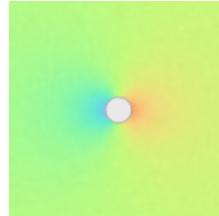
39559px



39559px



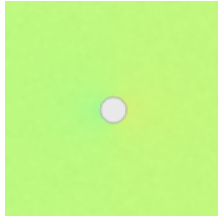
39559px



39559px



39559px

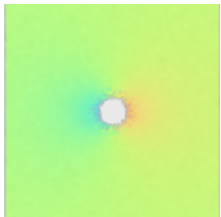


Reference Mode — Before vs After Fix (Noisy Data, $\sigma=2.0$)

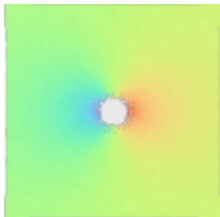
R=16
39197px



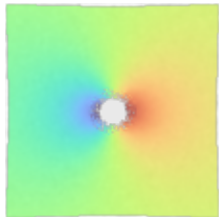
R=20
38661px



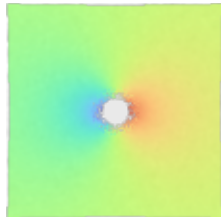
R=25
38004px



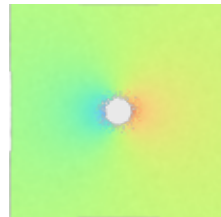
R=30
37003px



R=26
37694px



R=22
38359px



R=18
38765px



R=14
39473px



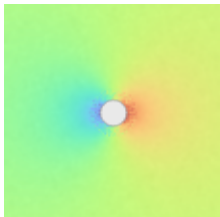
39559px



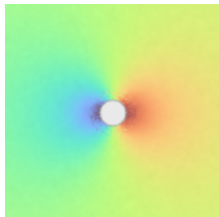
39559px



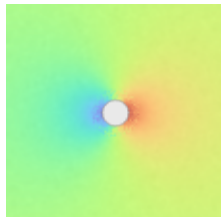
39559px



39559px



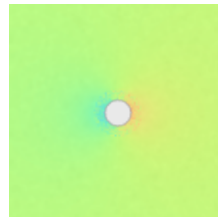
39559px



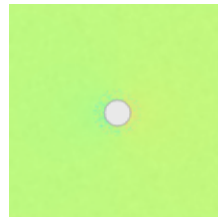
39559px



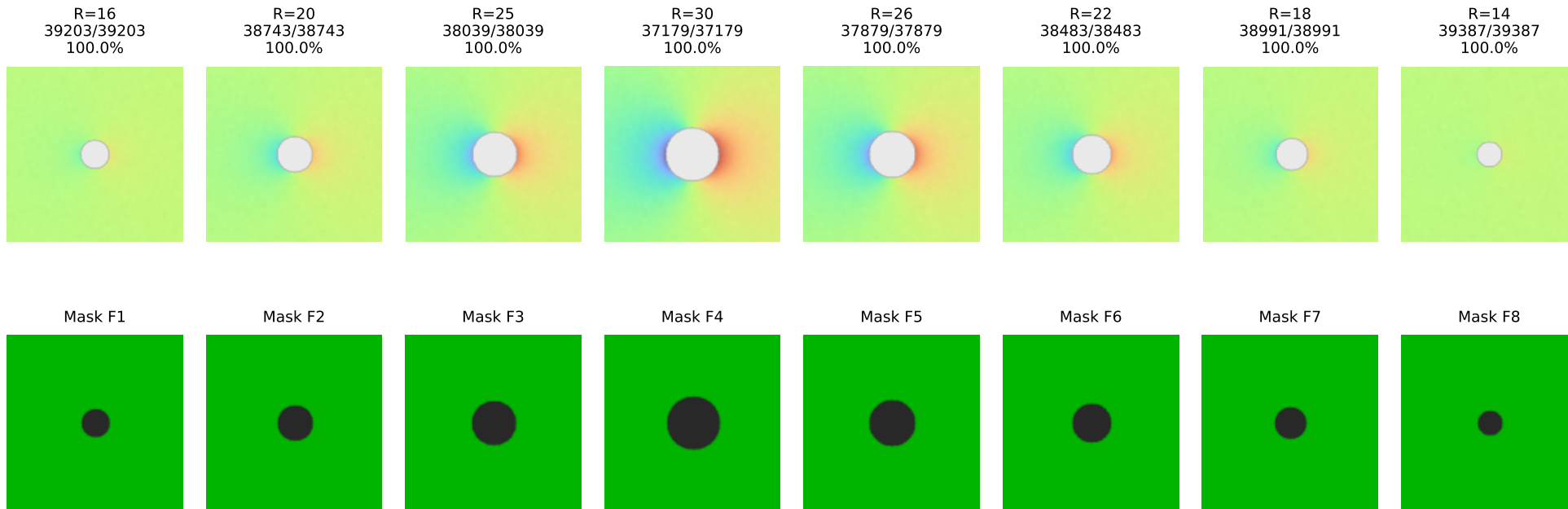
39559px



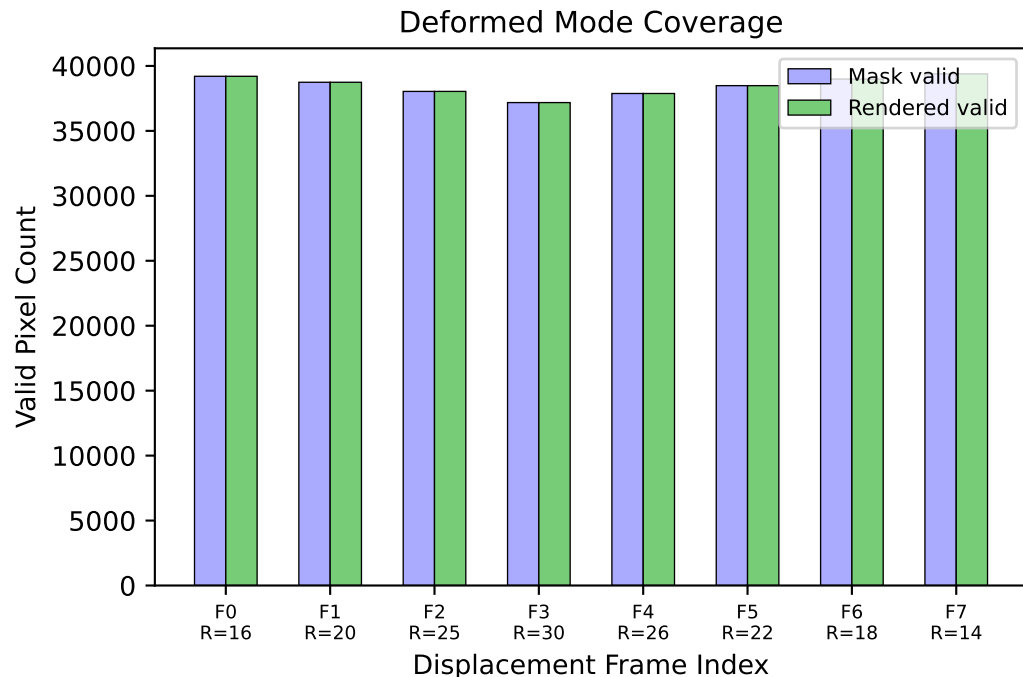
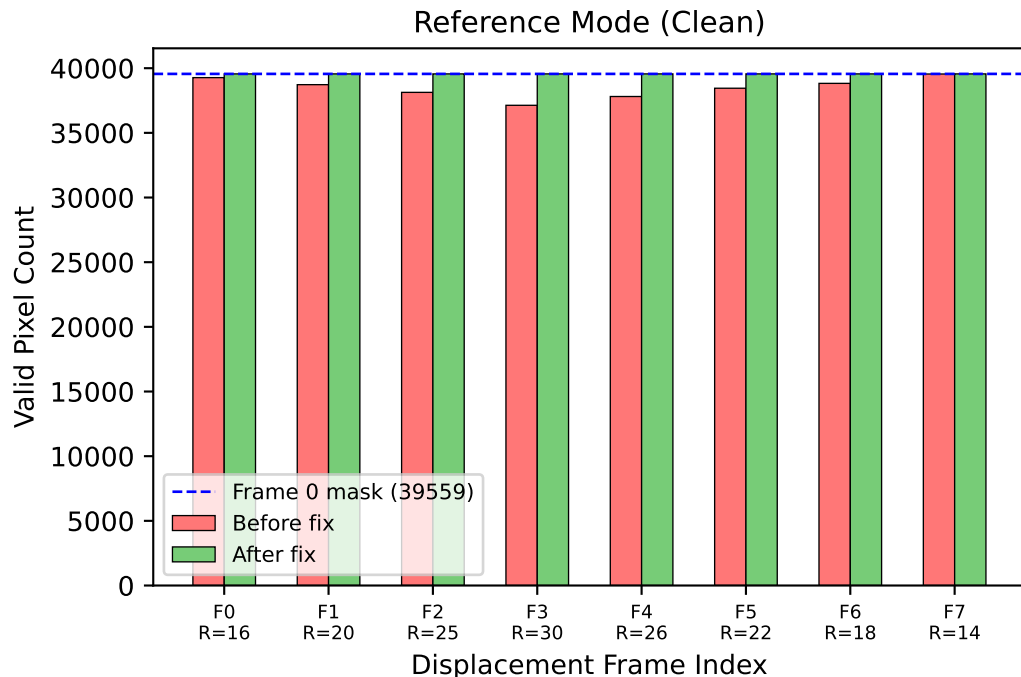
39559px



Deformed Mode — ROI Follows Per-Frame Mask (Growth + Shrink)



Pixel Count Comparison Across Frames



Test Suite Results Summary

All Tests: 510 passed, 0 failed

Bubble Visualization Tests (10 new tests)

test_bubble_visualization.py::TestReferenceMode

[PASS] test_valid_pixel_count_constant – All frames = 39559 px

[PASS] test_nan_pattern_matches_frame0 – 0 pixel mismatch for all frames

test_bubble_visualization.py::TestDeformedMode

[PASS] test_deformed_roi_follows_mask_growth – Valid area decreases with bubble

[PASS] test_deformed_roi_follows_mask_shrink – Valid area increases as bubble shrinks

[PASS] test_deformed_valid_inside_mask – 0 pixels rendered outside mask

test_bubble_e2e_visual.py::TestBubbleVisualE2E

[PASS] test_clean_incompressible – Ref mode constant: [39559, 39559, 39559, 39559, 39559, 39559, 39559, 39559]

[PASS] test_noisy_displacement – Ref mode constant despite 2997 NaN pixels

[PASS] test_deformed_mode_growth_shrink – All frames coverage = 100%

[PASS] test_deformed_mode_coverage_vs_mask – Coverage > 85% for all frames

Existing Tests (500 tests)

[PASS] All existing tests pass – no regressions

Files Modified

server/render_utils.py	– Added fill_mask_nan_gaps() function
server/routes/displacement.py	– Reference mode: fill NaN gaps (render + download)
server/routes/strain.py	– Reference mode: fill NaN gaps
server/deformed_warp.py	– Bounding box expansion + has_user_mask flag

Frontend Build

[PASS] Vite build successful (2587 modules, 5.29s)